The shooting method in the study of elliptic nonlinear equations

Valentino Badalucco

May 2026

Introduction to the Shooting Method

The shooting method reduces a Boundary Value Problem (BVP) to an Initial Value Problem (IVP).

Core Mechanism:

- Parameterize the solution by its initial amplitude $\alpha = u(0)$.
- Track the trajectory forward in space as an initial value problem.
- Tune α until the resulting trajectory satisfies the required properties and boundary conditions.

The Problem: From PDE to ODE

We seek stationary states for nonlinear dispersive equations, such as the nonlinear Schrödinger equation:

Dispersive Equation

$$i\partial_t \phi + \Delta \phi + |\phi|^{p-1}\phi = 0$$

Inserting the standing wave ansatz $\phi(t,x) = e^{i\omega t}u(x)$ yields the semi-linear elliptic ODE:

Stationary ODE

$$\Delta u - \omega u + |u|^{p-1}u = 0$$

1D reduction and conjecture

Assuming radial symmetry $u(x) = u(r)$ where $r = |x|$, the Laplacian simplifies, yielding our fundamental model:

Radial ODE

$$u'' + \frac{n-1}{r}u' + f(u) = 0, \quad u'(0) = 0, \quad u(0) = \alpha > 0 \ .$$

A natural question is:

Conjecture (Berestycki–Lions, 1983)

For all $m \geq 0$, does there exist a radial solution of the hereabove problem with exactly m zeros, such that the limit at infinity is 0? If so, is it unique?

Existence

The first part of the conjecture was solved shortly after

Theorem (McLeod–Troy–Weissler, 1990)

For any $m \geq 0$, there exists a radial solution of the aforementioned problem problem with exactly m zeros, such that the limit at infinity is 0.

The proof used in this paper is the blueprint for our work.

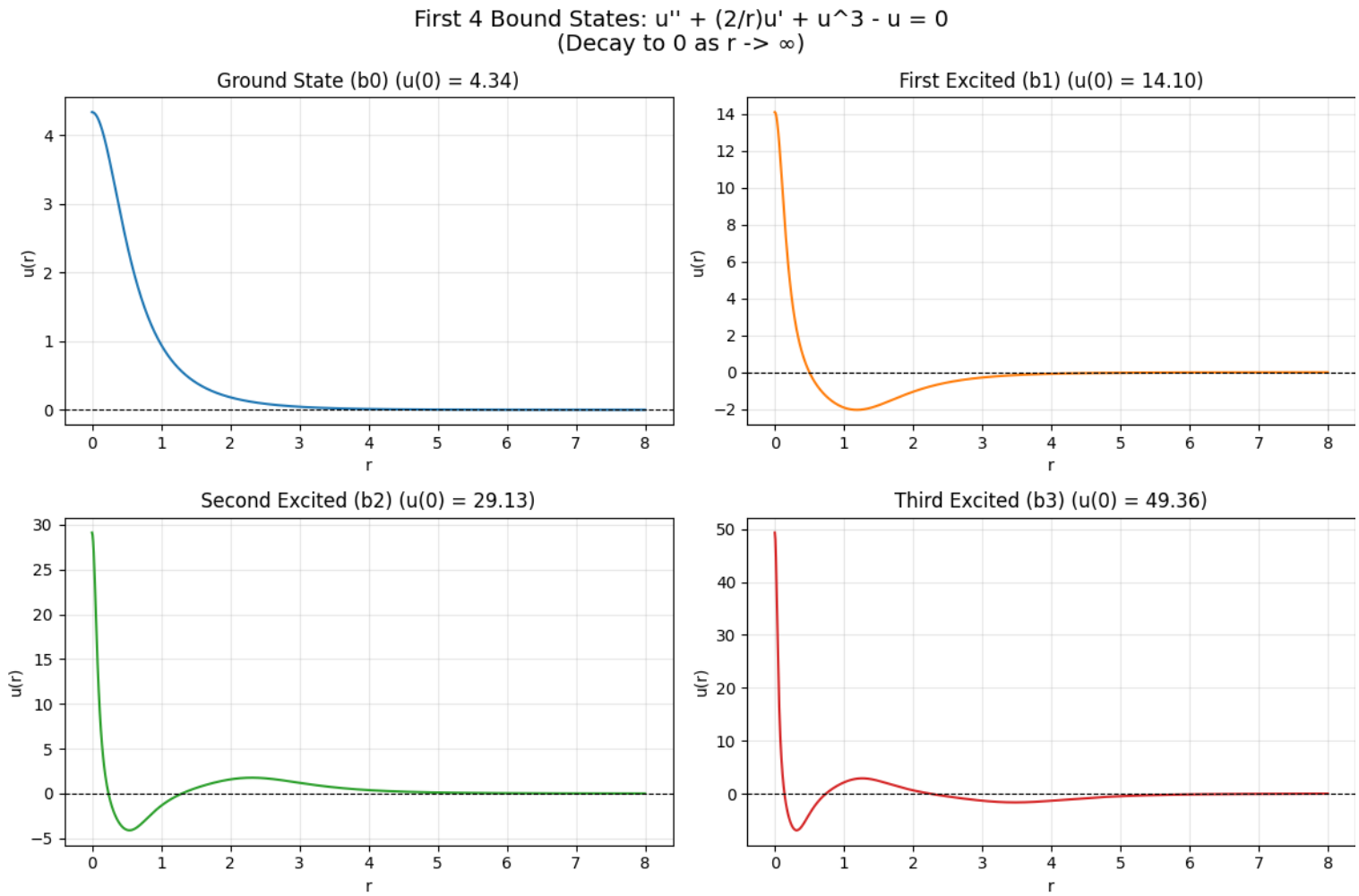
Why it makes sense

The method is grounded in the continuous dependence of the solution $u(r, \alpha)$ on the initial condition α.

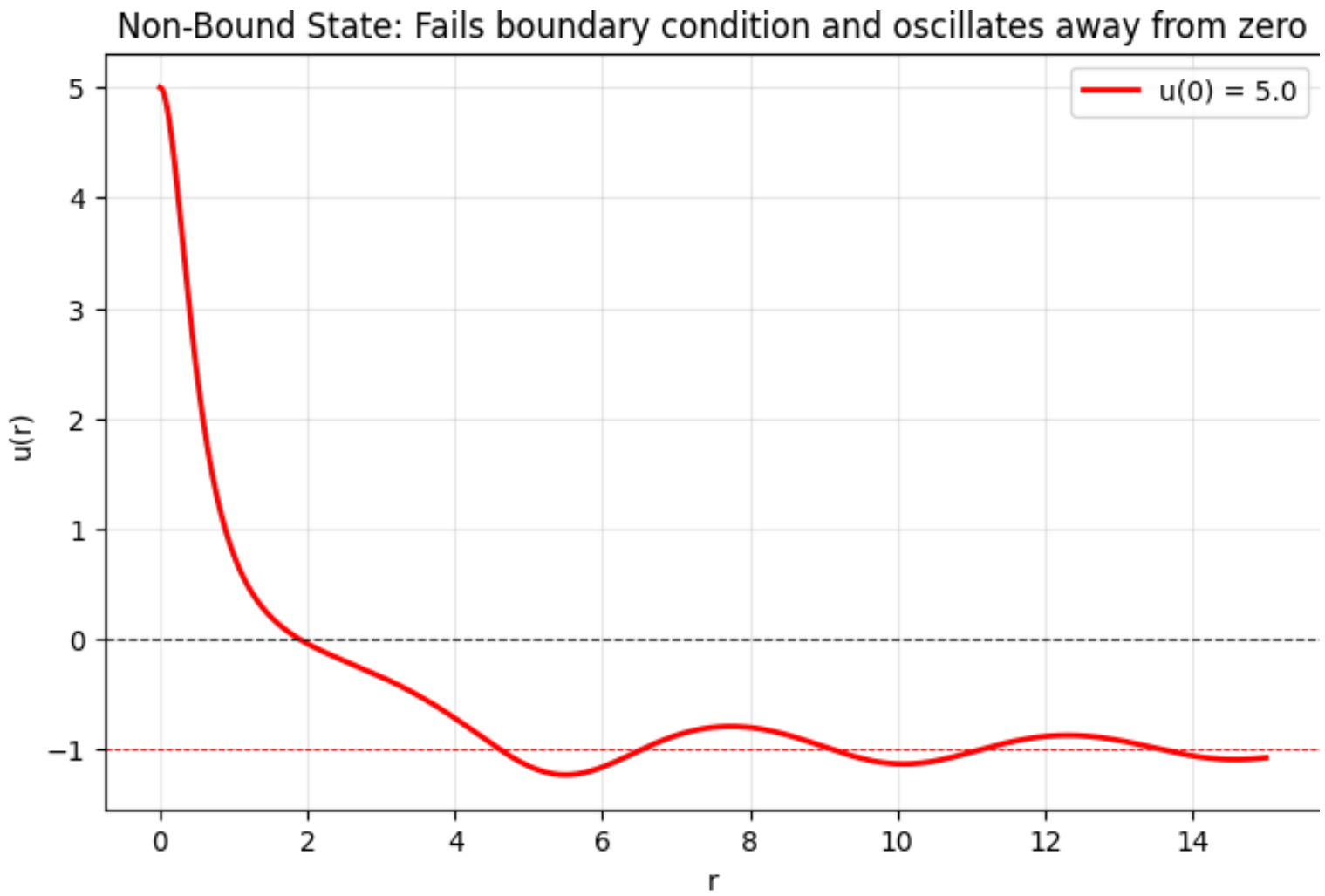
We expect highly oscillatory behavior for large α because the equation admits a well-defined scaled limit, and we can analytically prove this limit possesses infinitely many zeros.

We can also simulate the equation numerically to confirm whether this corresponds to reality, at least for some values of α, verifying the existence of distinct bounded states and oscillatory regimes.

Ground state and first three excited states



A counterexample



A Generalized Radial Problem

We can abstract away from strict geometric constraints entirely by replacing the Euclidean friction $\frac{n-1}{r}$ with a generic damping $g(r)$:

Generalized Radial ODE

$$u'' + g(r)u' + f(u) = 0, \quad u'(0) = 0, \quad u(0) = \alpha > 0$$

To preserve the core mechanics of the shooting method, the generalized friction must continuously drain energy from the system.

- **Strict Dissipation**: We assume $g(r) > 0$ for all $r > 0$. This ensures the energy functional $E(r) = \frac{1}{2}(u')^2 + F(u)$ is strictly decreasing along any trajectory.

A Generalized Existence Theorem

Beyond strict positivity, $g(r)$ must satisfy specific local and asymptotic structural conditions:

- **Pole Regularity**: $\lim_{r\to 0} rg(r) = \gamma \geq 1$. The parameter γ acts as an effective spatial dimension near the origin.
- **Generalized Subcriticality**: The nonlinearity $f(u) = \mathcal{O}(u^p)$ is restricted by an effective critical exponent: $1 < p < \frac{\gamma+3}{\gamma-1}$.
- **Integral regularity at Infinity**: the function $g(r)e^{2G(r)}$ is definitely non-decreasing (where $G' = g$).

Theorem (Bound states with prescribed nodes)

Under these generalized hypotheses, for every integer $k \geq 0$, there exists at least one initial amplitude yielding a bound state with exactly k nodes.

Translating the ODE to the Shooting Method

A rigorous execution of the shooting method requires five main steps:

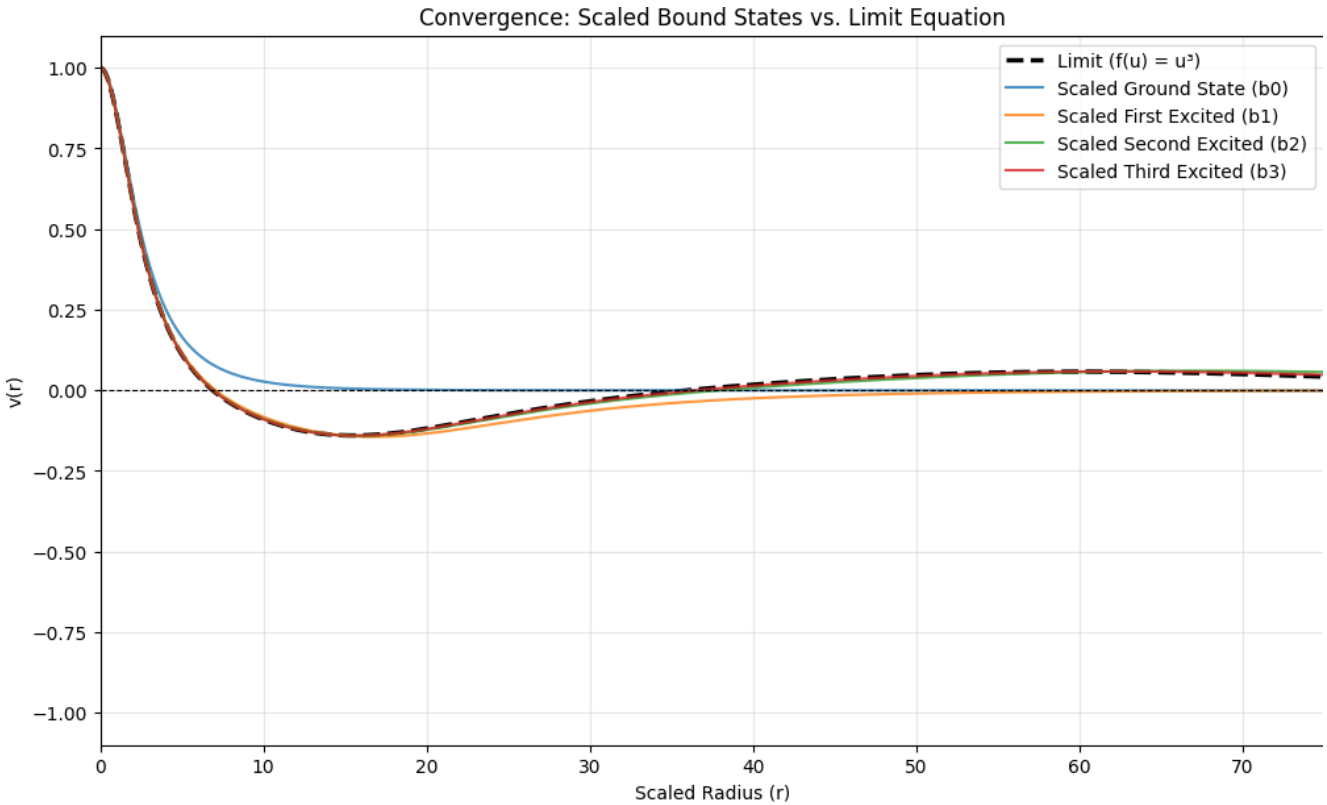
1. **Well-posedness**: Define the IVP with $u(0) = \alpha$, and ensure it is well-posed. Define $N(\alpha)$ as the number of zeros of $u(\cdot, \alpha)$.
2. **Small initial values**: Prove that for sufficiently small α, the solution never crosses the horizontal axis ($N(\alpha) = 0$).
3. **Controlled jumps**: Prove that as α varies continuously, $N(\alpha)$ never jumps upward by more than one at a time.
4. **Infinite oscillations**: Prove that $\lim_{\alpha \to \infty} N(\alpha) = \infty$, implying there are countably many jumps. This relies on the scaled limit having infinite zeros.
5. **Selecting the bound state**: Define the set $I_k = \{\alpha : N(\alpha) = k\}$. Deduce by energy methods that $\sup I_k$ yields a valid bound state.

Convergence to a limit

By scaling the parameters appropriately:

Rescaled Limit

$$v(r,\lambda) = \lambda^{-2/(p-1)} u\left(\frac{r}{\lambda}, \lambda^{2/(p-1)}\right) \xrightarrow{\mathcal{C}^2_{\text{loc}}} w(r)$$



Problem Formulation on Manifolds

Consider a spherically symmetric $\mathcal{C}^2$ manifold equipped with the warped product metric:

$$ds^2 = dr^2 + \psi(r)^2 dS^2_{n-1}$$

For a radial function $u(x) = u(r)$, the Laplace-Beltrami operator reduces the stationary equation $\Delta_s u + f(u) = 0$ to the generalized ODE:

$$u'' + g(r)u' + f(u) = 0, \quad \text{where } g(r) = (n-1)\frac{\psi'(r)}{\psi(r)}$$

We can then apply the previous result:

Theorem (Existence of Bound States on Manifolds)

Under the hypotheses above, the equation $\Delta_s u + f(u) = 0$, for every integer $k \geq 0$, admits at least one initial amplitude yielding a bound state with precisely k nodes.

Future Perspectives: Geometric Singularities

Our generalized existence theorem opens the door to studying PDEs on spaces with less regular geometries.

- **Regular Poles**: For any C^2 manifold, local flatness at the origin strictly forces the damping parameter to be $\gamma = \lim_{r \to 0} rg(r) = n - 1$.
- **Singular Poles**: Our generalized result relaxes this geometric constraint to any $\gamma \geq 1$.

Open Directions

This flexibility naturally accommodates manifolds with **controlled metric singularities** at the origin (such as conical points or sharp cusps).
Exploring the uniqueness and nodal structure of bound states on these generalized spaces is now possible

Uniqueness of Bound States

The shooting method guarantees existence, meaning $\sup I_k$ produces a bound state. However, it is not trivially the *only* choice. Proving uniqueness requires showing that $N(\alpha)$ never decreases, and that bound states only occur at jumps of $N(\alpha)$.

This is achieved by analyzing the variation function:

The Variation Function

$$v(r,\alpha) = \frac{\partial u}{\partial \alpha}(r,\alpha)$$

By understanding the shape and nodes of v, one proves that the zeros of u shift monotonically toward the origin as α increases. This interlacing guarantees $N(\alpha)$ never drops, and analyzing the divergent tail of v at infinity confirms $N(\alpha)$ strictly jumps exactly at bound states.

Results of a numerical simulation

By tracking the final rest state of the solution as a function of the initial amplitude, we observe a step-like structure characterizing the solution space.

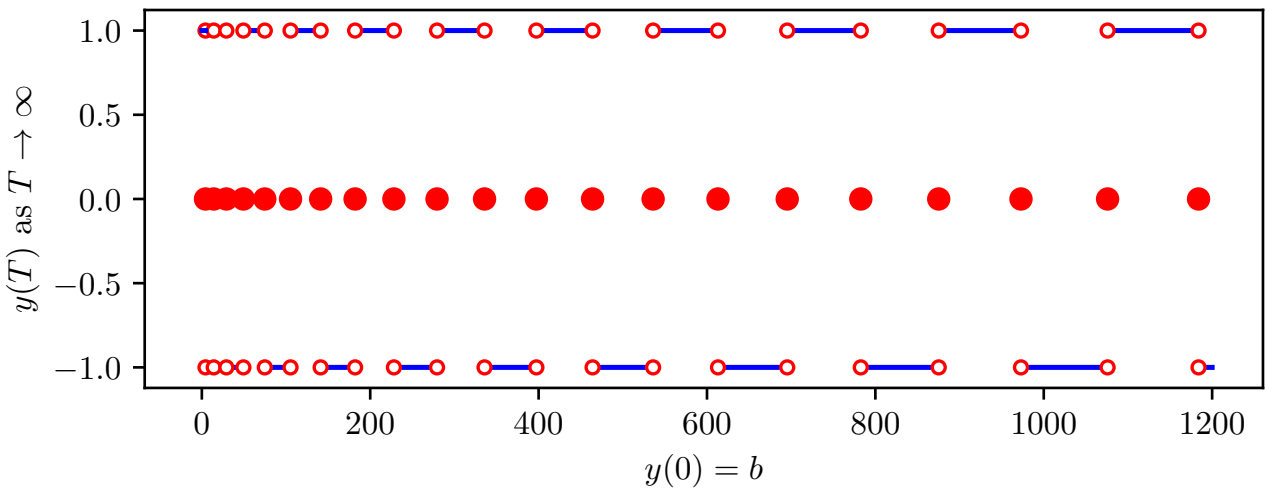


Figure: **Cohen–Li–Schlag (2024, Fig. 2).**

The red dots correspond to the states we are looking for.

A Uniqueness Theorem

In 2026, Tang provided a complete, rigorous proof of this uniqueness property for the classical Euclidean case, settling the conjecture raised by Berestycki and Lions in 1983.

Theorem (Tang, 2026)

For the model initial value problem

$$u'' + \frac{n-1}{r}u' - u + |u|^{p-1}u = 0, \quad u'(0) = 0, \quad u(0) = \alpha > 0$$

with $n \geq 3$ and $1 < p < \frac{n+2}{n-2}$, there exists a strictly increasing sequence of initial data $\alpha_0 < \alpha_1 < \alpha_2 < \dots$ diverging to infinity such that:

- $u(r, \alpha_0)$ is the unique ground state.
- $u(r, \alpha_k)$ is the unique k-node bound state for every integer $k \geq 1$.

Thank you!

Thank you!

Bibliography I



H. Berestycki and P. L. Lions.
Non-linear scalar field equations I, Existence of a ground state; II, Existence of infinitely many solutions.
Arch. Rational Mech. Anal., 82:313–375, 1983.



A. Cohen, Z. Li, and W. Schlag.
Uniqueness of excited states to $-\Delta u + u - u^3 = 0$ in three dimensions.
Analysis & PDE, 17(6):1887–1906, 2024.



C. Jones and T. Küpper.
On the infinitely many solutions of a semilinear elliptic equation.
SIAM J. Math. Anal., 17:803–835, 1987.

Bibliography II



K. McLeod, W. C. Troy, and F. B. Weissler.
Radial solutions of $\Delta u + f(u) = 0$ with prescribed numbers of zeros.
J. Diff. Eq., 83(2):368–378, 1990.



M. Tang.
Uniqueness of bound states to $\Delta u - u + |u|^{p-1}u = 0$ in $\mathbb{R}^n, n \geq 3$.
Invent. math., 243:245–291, 2026.